Evaluating Attendance at Wellspring

Registration system, age, and cancer type

Executive summary

Registration system
78% → 57%
First-month attendance was lower after implementation, among members with records.

Age
0.13 points / year
Age explains only **0.4%** of attendance variation.

Cancer type
Wilcoxon p = 0.303
Pooled comparison: not statistically significant; the rare groups are small.

Priority: Improve the completeness and comparability of registration and attendance records before drawing causal conclusions or targeting outreach by demographics.

Context and data: Measuring member participation

Wellspring provides support programs and services for people affected by cancer. It introduced a new registration system on **March 4, 2024** and expanded online programming.

Questions

1. Did first-month attendance differ before versus after implementation?
2. Does age predict attendance rate?
3. Do the most and least prevalent cancer types have different attendance rates?

Data used

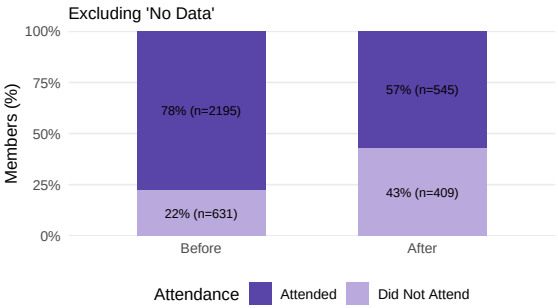
- Member file: member ID, age, cancer type, registration month
- Service file: member ID, event date, attendance status
- Service attendance rate = Present events ÷ recorded events

Observational analysis: the results describe associations, not the causal impact of the system.

RQ1: First-month attendance was lower after implementation

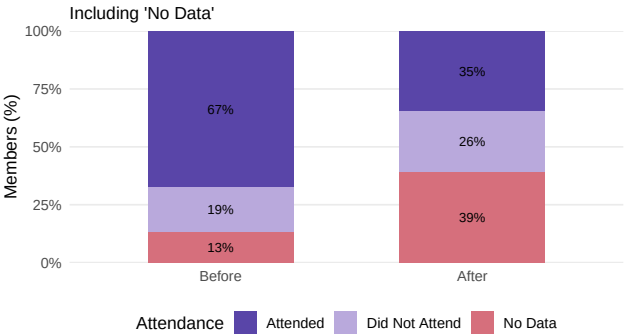
Question: Did the share attending at least one event within 30 days differ across implementation groups?

Method: Two-sample proportion test; members with **No Data excluded** from the primary comparison.



Before: 78% attended
After: 57% attended
Difference: 20.5 percentage points
95% CI: 17.0 to 24.0 points
$\chi^2 = 150.93$; $p < 0.001$

RQ1: Missing records complicate the before/after comparison



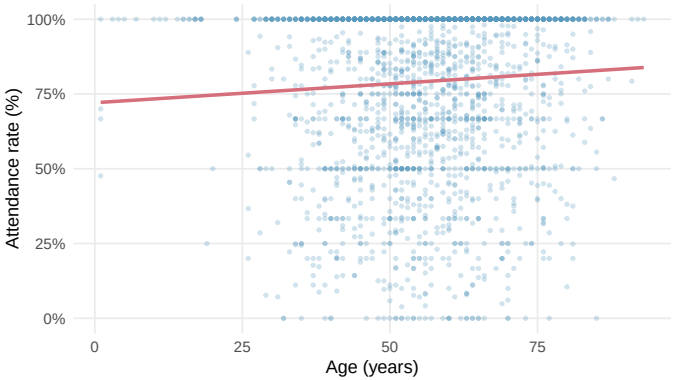
No Data: 13% before versus 39% after.

- After-period members have shorter potential follow-up.
- Missing or delayed service records may affect the comparison.
- Dates are **month-level**; everyone registered in March 2024 is counted as **Before**.

Interpretation: The observed decline does **not** establish that the new system caused it.

RQ2: Age is statistically associated with attendance, but weakly

Method: Member-level linear regression of service attendance rate (%) on age (years).

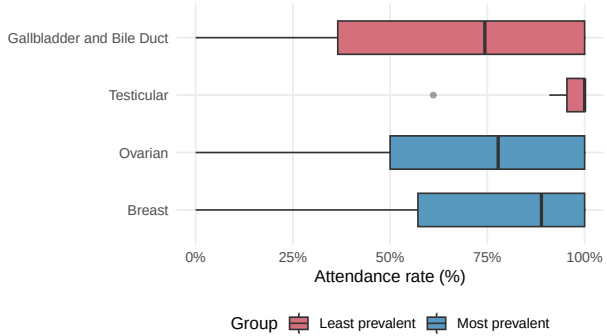


Each additional year: +0.13 percentage points (p = 0.002).
10 years: +1.3 points.
R^2**:** 0.0037 (about 0.4% of variation).
Sample: 2484 of 3779 members with service records.
Missing age: 1274; age 0: 20.

Age alone is too weak a predictor to justify age-based outreach.

RQ3: Cancer-type differences need cautious interpretation

Method: Compare two most versus two least prevalent types using member attendance rates; Wilcoxon rank-sum test on pooled groups.



Cancer type	n	Mean	Median
Breast	1491	76.7%	88.9%
Ovarian	208	69.0%	77.8%
Testicular	7	93.1%	100.0%
Gallbladder/Bile Duct	4	62.2%	74.4%

Wilcoxon $p = 0.303$ — not significant at 5%. Least-prevalent groups: **11 observations combined**.

Differences in sample means are descriptive; they do not establish significant pairwise differences.

Limitations

1. **Month level dates.** Registration days are unavailable. March 2024 registrations are all classified Before, although the change happened March 4.
2. **Missing and unequal follow-up.** No Data is more common in the After group, which also has less time for attendance to be observed.
3. **Incomplete age records.** 1274 of 3779 members with recorded services lack age (33.7%); 20 have age 0.
4. **Small cancer subgroups and observational design.** Only 11 observations are in the two least prevalent groups; unmeasured differences may affect every comparison.

Recommendations: Improve evidence before targeting outreach

1 Highest priority — Audit registration and service records.

Investigate missing attendance statuses, delayed entries, and whether post-implementation members have adequate follow-up. Preserve exact registration dates where possible.

2 Next — Monitor comparable cohorts.

Track a consistent 30-day attendance outcome only after each new member has had enough follow-up. Repeat the before/after analysis once the data are more complete.

3 Then — Investigate participation barriers.

Explore scheduling, accessibility, program preferences, and reminder processes. Do not target by age alone or infer rare-cancer needs from tiny samples.

Conclusion

- **Registration:** First-month attendance differs before versus after implementation, but missing records and unequal observation windows prevent a causal claim.
- **Age:** A statistically detectable positive relationship has very little explanatory power in practice.
- **Cancer type:** Results are descriptive and the pooled Wilcoxon test does not show a statistically significant difference; rare groups cannot support strong recommendations.

Takeaway: Better tracking is the most actionable immediate step for Wellspring.

Appendix: Definitions, exclusions, and statistical checks

- **RQ1 outcome:** Attended at least one event marked `Present` during days 0–30 after the approximated registration date. Main comparison excludes `No Data` (no dated service record).
- **Implementation groups:** Before = registration date earlier than March 4, 2024. Dates use the **first day of each recorded month**, therefore March is Before.
- **RQ2/RQ3 attendance rate:** `100 × (# events marked Present / # recorded attendance statuses)` per member; excludes members with no service attendance statuses.
- **Age regression:** Ages missing, 0, or over 120 are excluded.
- **RQ3 categories:** Split semicolon-separated cancer types; exclude blank/Other; most/least prevalent chosen by counts among members with service records. Members with more than one type may appear multiple times.
- **Tests:** Two-sample proportion test (RQ1), OLS regression (RQ2), pooled Wilcoxon rank-sum test (RQ3). All statistics are computed when this file renders.